

From Reverse Entailment Diagnostics to a Universal Linguistic Domain

A submission-complete preprint with RED dataset examples, formal appendix, and neuro-computational extensions

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Abstract

This paper develops a rigorous theoretical transition from the **Reverse Entailment Diagnostic (RED)**—a bidirectional test for nominal potential—into a more general claim about the architecture of linguistic cognition: the existence of a **Universal Linguistic Domain (ULD)** conceived as a *background geometrical space* of position-types (“slots”) and their lawful transformations. RED begins as an operational tool: it tests whether linguistic material of diverse surface categories (roots, phrases, operators, interjections) can be coerced into **referential noun-slot behavior** via nominal derivation and reverse entailment. The cross-context stability of RED

outcomes motivates a *latent-geometry argument*: if heterogeneous material can systematically acquire stable noun-like addressability through a small class of operations, then the cognitive system must contain pre-existing **slot families** that license those operations, constrain interpretation, and structure interface mappings. I propose that ULD minimally contains five families of slot-types: **referential**, **predicational**, **operator**, **linking**, and **packaging**. I further situate **exeme** (a modality-neutral expression pattern with no commitment to word status, category, or meaning) as a micro-level representational instantiation of ULD: an exeme becomes a lexeme not by self-association but by **placement into a licensed ULD subdomain**, where it inherits syntactic behavior from slot constraints and interpretive content from constructional and discourse operators. The paper provides (i) a numbered RED example set with consistent interlinear conventions following the Leipzig Glossing Rules , (ii) a compact formal appendix defining ULD as a typed graph and mapping diagnostics-to-slot families, (iii) computational applications connecting RED/ULD to UD parsing and SRL resources and transformer-based models , and (iv) neurolinguistic predictions grounded in contemporary network models of language (temporo-frontal circuitry; unification architectures; ERP meaning signatures).

Keywords: Reverse Entailment Diagnostic; universal noun slot; Universal Linguistic Domain; nominalization; exeme; slot families; construction grammar; distributed morphology; UD parsing; semantic role labeling; neurolinguistics.

1. Introduction

A recurring methodological tension in linguistics is the gap between **diagnostics** and **ontology**. Diagnostics are local tests: they reveal whether a form behaves like a noun, a verb, a determiner, a complementizer, etc., under particular conditions. Ontology is global: it concerns what kinds of cognitive objects and position-types must exist so that such diagnostics are even meaningful and cross-linguistically stable.

This paper addresses that tension in the context of the **Reverse Entailment Diagnostic (RED)**

(Author 2025a), originally introduced to test a specific hypothesis: that a **universal noun-slot** exists as a stable cognitive-syntactic region that can host heterogeneous linguistic material once nominalization operations and discourse addressability are satisfied. The present goal is to develop a submission-ready argument that **RED entails a broader architecture**, namely a **Universal Linguistic Domain (ULD)** (Author 2025b): a background “geometrical” space of slot families that structure categorization, composition, and interpretive inheritance.

The central claim is a *route of necessity*:

1. **RED is not merely a trick for making things look like nouns.** It functions as an operational probe of a latent property: *whether the system can treat an item as a referential, discourse-addressable entity*.
2. **If RED generalizes across diverse bases and languages**, it reveals a stable position-type in cognition: a **referential slot** that is not reducible to the surface category of the base.
3. **If one slot-type is inferable as a latent constant**, then other position-types must also exist (predication, operators, linking, packaging), because reference alone does not yield full linguistic computation.
4. Therefore, **ULD is a necessary background space** for the interpretation and stability of RED itself.

In addition, this paper integrates my refinement: **exeme does not associate itself with a lexeme; it becomes a lexeme by placement**. On this view, exeme is a modality-neutral expression pattern that becomes typed through insertion into ULD subdomains (Author 2025c). RED serves as a bridge from empirical diagnostics to the postulation of ULD.

2. Background and related work

The RED→ULD route resonates with, but is not identical to, several established lines of theory:

2.1 Category flexibility and category-neutral lexical material

Lexicalist and post-lexicalist traditions have long noted that a single underlying item can appear with different category features depending on structural context. Chomsky's discussion of "neutral" lexical entries (free with respect to category features, with distributional generalizations captured in the lexicon) anticipates the idea that categorization is not inherent to the base form alone.

Distributed Morphology (DM) makes this central: roots are category-neutral and acquire nominal/verbal/adjectival status via functional structure. The RED/ULD program differs in emphasis: it treats **slot families** as a cognitive geometry, and uses a diagnostic (RED) to infer the existence and stability of those slot families.

2.2 Slot-like positions and constructional licensing

Construction Grammar argues that syntactic patterns carry meaning and impose constraints independent of individual words. This aligns with ULD's core idea: position-types constrain what can appear there and what interpretive commitments follow from placement.

2.3 Linking and argument structure

Argument selection and thematic mapping are classically studied through proto-role entailments and alternation behavior. ULD's **linking slots** generalize this: the mapping between semantic roles and syntactic positions is treated as a region of the domain with its own constraints and operators.

2.4 Formal semantics and operator structure

Formal semantics emphasizes that operators (quantification, negation, tense/aspect/modality) systematically compose with predication material. Standard treatments assume typed composition and scope management as part of grammar. ULD treats operators as a distinct slot family because they impose non-local interpretive effects (scope, presupposition, modality) not reducible to lexical meaning.

2.5 Neurocognitive architectures

Contemporary neurobiological models increasingly emphasize distributed networks (temporal cortex, inferior frontal cortex) and **unification** operations over retrieved memory items. Meaning-related ERP signatures (e.g., N400) provide an additional interface-level measure of how slot placement and composition modulate semantic processing.

3. The Reverse Entailment Diagnostic (RED)

3.1 RED as a bidirectional test (operational definition)

RED is defined here as a *three-stage diagnostic battery* for **nominal potential**:

1. **Nominal derivation:** Apply a nominalizing operation (possessive/genitive marker, nominalizing morphology, or a construction that yields nominal behavior).
2. **Reverse entailment:** If the derived form is interpretable/acceptable, infer backwards that the base has latent nominal potential (even if not canonically nominal).
3. **Noun-slot verification:** Place the base (or its minimally derived counterpart) into standard noun-slot environments (subject, object, determiner positions; discourse anaphora) to confirm entity-like behavior.

This diagnostic does not assume that “nounhood” is a fixed property of the base. Instead, nounhood emerges as a **licensed mode of addressability** when the relevant operations and slot constraints are satisfied.

3.2 What RED measures (and what it does not)

RED measures a *capacity of the grammar-cognition system*: the ability to treat material as a **referential object**. It does **not** claim that all bases are “really nouns,” nor that derivation collapses lexical distinctions. Rather, RED isolates the **existence of a stable referential region** that can host diverse material once the system is permitted to package it as an entity.

3.3 Why a diagnostic implies a background space

A diagnostic is only meaningful if:

- there exist **stable target positions** (slots) that define what counts as success; and
- the operations used in the diagnostic correspond to **lawful transformations** in the grammar.

Thus, if RED is robust and general, it presupposes a structured background—precisely what ULD formalizes.

4. The RED → ULD transition: a rigorous route

This section is the core theoretical bridge.

4.1 Step 1: From repeated diagnostic success to a latent position-type

Assume the following empirical generalization (Author 2025a):

Across languages, heterogeneous bases can be coerced into noun-like behavior via a small family of nominalizing/possessive/genitive operations, and the resulting forms are discourse-addressable and noun-slot compatible.

If this generalization holds beyond isolated examples—i.e., if it is stable across constructions, lexical classes, and languages—then it is not parsimonious to treat each success as an idiosyncratic trick. The most economical explanation is that the cognitive grammar includes a **latent position-type** that remains constant: a **referential slot**.

This inference is structurally analogous to inferring a latent variable from a measurement battery in psychometrics: the diagnostic's convergence suggests the existence of a stable underlying dimension.

4.2 Step 2: From one slot-type to a domain of slot families

A single referential slot is not enough to generate full grammar. If linguistic computation were only reference, we could name entities but not attribute properties, compose propositions, or manage scope.

Therefore, once we acknowledge one stable position-type (referential), we must ask what minimal additional position-types are required for a functioning linguistic system. This yields the argument for **ULD**:

- **Predicational slots:** to attribute properties, build event structure, and form propositions.
- **Operator slots:** to manage determiners, quantification, tense/aspect/modality, negation, and scope.
- **Linking slots:** to map semantic roles to syntactic positions and organize argument structure.
- **Packaging slots:** to construct larger units (nominalization, relativization, subordination, coordination) and embed structures.

The point is not to impose an English-centric template, but to define **families** of slot types that are required by minimal communicative computation: reference + predication + operator control + linking + packaging.

4.3 Step 3: ULD as “background geometry,” not a list of categories

ULD is not a universal inventory of parts of speech. It is a **space of position-types** with constraints and operations. Parts of speech are **surface stabilizations** of repeated placements into regions of the domain.

Formally (preview; see Appendix B), we treat ULD as a typed graph:

- nodes = slot-types (positions);
- edges = licensed operations/transitions (nominalization, predication, operator attachment, argument linking, packaging);
- typing = constraints that regulate what can occupy a node and what interpretation follows.

On this view:

- **RED identifies a specific region** of ULD (the referential noun-slot family).
- ULD generalizes the logic of RED to the full set of slot families needed for grammar.

4.4 Step 4: Why ULD must be universal (in the relevant sense)

“Universal” here does not mean “identical surface expression in all languages.” It means:

1. Slot families exist as **necessary functional regions** of the cognitive-linguistic system.
2. Languages may realize these regions through different morphosyntactic means, but the **computational roles** remain stable.

This is consistent with cross-linguistically consistent annotation goals in computational linguistics, where universal POS/features and dependency relations are proposed as stable comparative

anchors while allowing language-specific realizations.

5. Exeme within the RED–ULD architecture

5.1 Exeme as micro-model of ULD

I propose **exeme** as a modality-neutral expression pattern with no commitment to word status, grammatical category, or meaning (Author 2025c). In the RED→ULD architecture, exeme functions as a **micro-level representational object** whose interpretive destiny is determined by placement.

5.2 Correcting the directionality: lexemization-by-placement

My correction is adopted as a definitional constraint:

An exeme does not associate itself with a lexeme.

It becomes a lexeme by being **placed into a licensed slot** in a ULD subdomain, under a construction that stabilizes category and meaning.

Thus, lexemehood is not an intrinsic property of the exeme; it is an *emergent stabilization* produced by repeated successful placement and discourse reuse.

5.3 RED as the bridge between exeme and ULD

RED provides an operational route:

- start with minimally typed material (exeme-like or category-flexible material);
 - apply nominal packaging;
 - test whether the system treats it as a stable referential entity;
 - infer the existence and constraints of the referential region of ULD.
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6. Data and diagnostics: numbered examples with glossing conventions

6.1 Conventions

Examples follow standard interlinear formatting conventions per the Leipzig Glossing Rules.

Abbreviations used here (see Appendix A): **GEN** genitive, **POSS** possessive, **NMLZ** nominalizer, **COP** copula, **DET** determiner, **1SG/3SG** person, **F** feminine, **TOP** topic.

Note on status: The examples below are presented as *diagnostic illustrations* aligned with the RED dataset and are formatted for consistency; speaker-validation and corpus attestation should be added for journal submission where required (see §10).

6.2 Example set 1: Nominal packaging of category-flexible bases (“square/quadrilateral”)

(1) English (base ↔ nominal-packaged)

a. *Base (NP/AP flexible):*

(1a) **square**

b. *Derived (POSS-marked nominal):*

(1b) **square**=’s side-s

square=POSS side-PL

‘the square’s sides’

(2) Mandarin Chinese (GEN/DE packaging)

(2) zhèngfāngxíng de biān

square GEN side

‘the square’s sides’

6.3 Example set 2: Nominal packaging of a VP (“to book carefully”)

(3) English (gerund + possessive packaging)

a. *Base (VP/TP environment):*

(3a) My job is **to book carefully**.

b. *Derived (nominalized event + POSS):*

(3b) My job is **booking carefully**=’s advantage.

booking careful=POSS advantage

‘the advantage of booking carefully’

(4) Mandarin Chinese (DE packaging)

a. *Base:*

(4a) wǒ xǐhuān xiǎoxīn yùdìng

1SG like careful reserve

'I like to book/reserve carefully.'

b. *Derived:*

(4b) xiǎoxīn yùdìng de yōudiǎn

careful reserve GEN advantage

'the advantage of booking carefully'

6.4 Example set 3: Possessive NP internal packaging

("her house")

(5) English

(5) **her** house='s door

3SG.F house=POSS door

'the door of her house'

(6) Mandarin Chinese

(6) tā de fángzi de mén

3SG GEN house GEN door

'the door of her house'

(7) Tongan (POSS system; diagnostic packaging)

(7) **hono** fale 'a e matapā

3SG.POSS house GEN DET door

'the door of her/his house'

(Gloss reflects the intended possessive/genitive packaging region; verify classifier choice and possessive series in speaker consultation for submission.)

(8) Spanish (de-genitive packaging)

(8) la puerta **de** su casa

DET door GEN 3SG.POSS house

‘the door of her/his house’

(9) Arabic (iḍāfa/genitive packaging; MSA-style)

(9) bāb-u bayt-i-hā

door-NOM house-GEN-3SG.F

‘the door of her house’

6.5 Example set 4: Packaging interjections as discourse entities (“oh/ah/’oi”)

(10) Spanish (nominal packaging of interjection)

(10) el significado **de** “¡oh!”

DET meaning GEN oh

‘the meaning of “oh!”’

(11) Mandarin Chinese (quotation/nominal reference)

(11) “āyà” **zhè** gè gǎntàn

ah.ya this CL exclamation

‘this exclamation “āyà” ’

(12) Tongan (nominal reference to interjection; diagnostic form)

(12) “oi” ’a e mālie

oi GEN DET beauty

‘the beauty/force of “oi”’

Diagnostic takeaway: Examples (1)–(12) instantiate the RED logic: nominal packaging

operations create **discourse-addressable entities** out of bases that are not canonically nouns (VPs, interjections), supporting the inference that a stable **referential slot family** exists as part of ULD.

7. Theory: deriving the five ULD slot families from RED

This section answers my question: “How do we move from ULD to these families—and from what theoretical aspect do we move to ULD? Exeme or RED?”

7.1 RED provides the empirical lever; exeme provides the micro-representation

- **RED** is the methodological lever that reveals the existence of stable slot behavior (referential noun-slot).
- **Exeme** is a representational hypothesis about the pre-typed or minimally typed units that can be placed into ULD.

So: **RED** motivates ULD, and **exeme** models how ULD can be instantiated at micro-scale.

7.2 Minimal decomposition argument (why these

families are required)

Once RED establishes a stable referential region, the rest follows from functional necessity:

1. **Referential slots** (entity tracking; discourse addressability): established by RED.
2. **Predicational slots** (property attribution; event structure): required to say something about referents.
3. **Operator slots** (det/quant/TAM/negation): required to manage scope, temporality, modality, polarity, definiteness.
4. **Linking slots** (argument structure/thematic mapping): required to map event participants to syntactic positions.
5. **Packaging slots** (nominalization, relativization, subordination, coordination): required to build complex expressions and allow recursion.

These are not “categories” but **regions in a structured domain**.

7.3 ULD as typed geometry

ULD’s “geometrical” character is captured by:

- **locality:** operations apply in defined neighborhoods (e.g., nominal packaging changes what the item can do);
 - **constraints:** not everything can occur everywhere;
 - **inheritance:** behavior emerges from placement (the key exeme claim).
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8. Computational applications

8.1 Why RED/ULD matter for NLP

Most NLP pipelines already assume something ULD-like: they distinguish words/phrases by POS, dependency roles, operator-like tokens, argument structure, and packaging constructions. Universal Dependencies explicitly organizes annotation into a morphological layer (lemmas, universal POS, features) and a syntactic layer relating predicates, arguments, and modifiers. This is structurally compatible with ULD slot families.

Similarly, PropBank-style SRL annotates predicate argument structure while remaining relatively agnostic about quantification and discourse structure. RED can enrich this by explicitly tagging when non-nominal material is coerced into referential packaging.

8.2 A RED-to-ULD computational mapping (pipeline sketch)

Inputs: sentence(s), language ID (or multilingual model), tokenizer, UD parser, optional SRL system.

Outputs: ULD slot assignments + RED flags + coercion edges.

1. **Parse and annotate:** obtain POS, dependencies, morphological features (UD).
2. **Detect nominal packaging candidates:** genitives/possessives, nominalizers, quoting constructions, determiners.

3. **Apply RED constraints:** if packaging yields a discourse-addressable nominal constituent, infer nominal potential for base.
4. **Assign ULD region:** map the constituent to **referential**, link to **operator** (determiners/quantifiers), connect to **linking** if SRL identifies participant roles.
5. **Learnable extension:** embed these diagnostic signals into transformer representations (feature prompting or auxiliary losses).

8.3 Integration with transformers

BERT-style models pretrain bidirectional contextual representations and can be fine-tuned for downstream tasks with minimal architecture change. A concrete ULD application is **multi-task supervision**:

- primary task: parsing or SRL;
- auxiliary task: predict ULD slot-family labels and RED-coercion edges.

This makes ULD not merely interpretive but computationally testable: if ULD captures stable generalizations, slot-family prediction should transfer across domains and languages better than surface POS alone.

9. Neuroscientific implications and predictions

9.1 ULD as a cognitive architecture hypothesis

ULD predicts that language processing requires:

- **stored building blocks** (lexical memory), and
- **structure-building/unification mechanisms** that place and combine them.

This matches modern frameworks emphasizing temporo-frontal networks and integration operations.

9.2 Slot-family correlates (hypothesis space)

1. **Referential slot operations (RED region):** should increase activation in networks associated with nominal/reference processing and discourse tracking, especially when non-nominal material is coerced into an entity.
2. **Predicational slots:** should recruit inferior frontal and temporal regions associated with syntactic structure building and semantic composition.
3. **Operator slots:** should show sensitivity to scope and control demands, consistent with broader control/unification components.
4. **Meaning integration signatures:** coercion into referential packaging should modulate semantic ERP components linked to meaning processing (e.g., N400 amplitude effects under expectation and integration load).

9.3 A specific RED prediction (testable)

Compare:

- **non-coerced VP:** "I like to book carefully."

- **coerced referential nominal:** “The advantage of booking carefully is...” / “Booking carefully’s advantage is...”

ULD predicts that the coerced form requires additional *packaging + referential-slot placement*, producing stronger integration/unification demands than the non-coerced VP, potentially reflected in temporo-frontal coupling (fMRI/MEG) and/or ERP integration signatures. This aligns with broader evidence that language functions depend on distributed networks rather than isolated modules.

10. Limitations and future work (submission-style)

10.1 Limitations

1. **Data status:** Several RED examples are diagnostic illustrations; submission should add speaker judgments, corpus attestations, and clearer language-specific morphological analyses (especially for possessive systems with classifier distinctions).
2. **Overgeneration risk:** If nominal packaging is too permissive, RED could misclassify pragmatically coerced but grammatically marginal items. A graded acceptability protocol is needed.
3. **Typological breadth:** The present dataset targets a small set of languages; a stronger universality claim should include families with radically different nominalization and reference strategies.
4. **Neuro claims are currently predictive, not confirmatory:** the proposed mapping from slot families to networks is grounded in existing architectures but requires direct experimental testing.

5. **Computational evaluation is prospective:** the pipeline sketch requires benchmark tasks, error analysis, and ablation studies against UD/SRL baselines.

10.2 Future work

1. **Expanded RED battery:** add determiner licensing, pronominalization/anaphora, and quantifier scope diagnostics as separate probes for referential vs operator slot families.
 2. **Formal completeness:** develop a full typed-graph semantics for ULD with explicit composition rules (Appendix B is a compact start).
 3. **Cross-ling dataset:** build a RED-annotated corpus with consistent glossing and judgment metadata.
 4. **Neuroexperimental program:** test RED-coercion contrasts with ERP/fMRI and connectivity analyses under controlled contexts.
 5. **NLP deployment:** implement ULD slot-family prediction as an auxiliary objective for multilingual transformers and evaluate transfer to low-resource languages.
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11. Conclusion

This paper provided a rigorous academic route from **RED** to **ULD**:

- RED is an operational diagnostic that reveals stable **referential slot behavior** across heterogeneous bases.
- Such stability presupposes a structured background space of position-types: **ULD**.
- Once a referential region is established, minimal linguistic computation forces additional slot

families: predication, operator, linking, and packaging.

- **Exeme** is best understood as a micro-level representation that becomes a lexeme by **placement into ULD**, not by self-association.
- The resulting framework is not only theoretical: it yields computational mappings to UD/SRL infrastructures and testable neurolinguistic predictions.

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Appendix A. Glossing abbreviations (compact)

1SG/2SG/3SG person-number; **F** feminine; **DET** determiner; **GEN** genitive; **POSS** possessive clitic/suffix; **NOM** nominative; **COP** copula; **CL** classifier; **TOP** topic; **NMLZ** nominalizer. Hyphens indicate morpheme boundaries; equals signs indicate clitics (Leipzig conventions).

Appendix B. Formal appendix: ULD as a typed graph (compact)

Let **ULD** be a typed directed graph:

- $G = \langle V, E, \tau, \kappa \rangle$
 - V : a set of *slot nodes* (position-types).
 - $E \subseteq V \times V$: directed *operation edges* (licensed transitions).
 - $\tau: V \rightarrow \mathbb{T}$: a typing function mapping each node to a slot-family type in $\mathbb{T} = \{\text{REF}, \text{PRED}, \text{OP}, \text{LINK}, \text{PACK}\}$.
 - $\kappa: E \rightarrow \mathcal{R}$: a constraint/compatibility function assigning well-formedness conditions and interpretive commitments to transitions.

Occupancy relation. For an expression token x (exeme-level object), define **Occ**(x, v) if and only if x is licensed to occupy slot v under the constraints κ of incoming edges and the local constructional environment.

Lexemization-by-placement. An exeme token x becomes a lexeme-token $\ell(x)$ relative to a discourse model D if there exists a slot v and a construction c such that:

- **Occ(x, v)** under **c**;
- **D** stores **x** as discourse-addressable (for REF slots) or compositionally stable (for other families);
- repeated successful placements converge to a stable mapping $\mathbf{x} \mapsto \ell$ (lexeme identity stabilization).

RED region (formalized). RED is a diagnostic over candidate edges entering **REF** nodes. It tests whether there exists a nominal packaging edge **ePACK: PACK** $\rightarrow$ **REF** such that the packaged form is interpretable and slot-licensed, and whether the base retains a recoverable mapping back to entity-like reference (reverse entailment).

Appendix C. Mapping diagnostics to slot families (compact)

Define diagnostic families $\Delta = \{\Delta\text{REF}, \Delta\text{PRED}, \Delta\text{OP}, \Delta\text{LINK}, \Delta\text{PACK}\}$ and a mapping **M**: $\Delta \rightarrow \mathbb{T}$.

- **ΔREF (referential):** RED nominal derivation + reverse entailment + noun-slot placement; determiner/pronominalization compatibility.
- **ΔPRED (predicational):** copular predication tests; event-structure licensing; aspectual compatibility.
- **ΔOP (operator):** scope diagnostics (negation/quantifiers), definiteness effects, TAM contrasts.
- **ΔLINK (linking):** alternations (active/passive), case/argument realization, SRL stability.

- **Δ PACK (packaging):** nominalization, relativization, subordination, coordination diagnostics (embedding and constituency).

The point of **M** is not to force one-to-one mappings in all languages, but to define a systematic way to infer which ULD region a construction is probing.